

Agentic AI
Assignment Alpha
Ibrahim Johar Farooqi
23K-0074
BA1-7A

Date _____ 20__

Q-1) 1280×720 px. w/ X bits of colors per information are used

1) value of X when bits per frame are 29,491,200

$$\begin{aligned} \rightarrow \text{pixels per frame} &= 1280 \times 720 \\ &= 921,600 \text{ pixels} \end{aligned}$$

$$\text{bits per frame} = \text{pixels} \times \text{bits per pixel}$$

$$29,491,200 = 921,600 \times \text{bits per pixel}$$

$$\text{bits per pixel} = \frac{29,491,200}{921,600}$$

$$= 32 \text{ bits per pixel}$$

2) bytes per frame

$$\text{bytes per frame} = \frac{29,491,200}{8}$$

$$= 3,686,400 \text{ bytes}$$

$$\approx 3.69 \text{ MB}$$

3) bits in one second of percept at 70 MB/s

$$1 \text{ MB} = 10^6 \text{ bytes}$$

$$70 \times 10^6 \times 8 = 560,000,000 \text{ bits [560 Mbps]}$$

$\rightarrow$ at 30 fps;

$$\frac{560,000,000}{30} = 18.67 \text{ Mbits per frame}$$

Q.2) "AI Governance emerging course"

AI:

Governance

Emerging

Course

$Q = [1.5, 3]$

$Q = [2, 4]$

$Q = [2.2, 2.4]$

$Q = [2.4, 4.4]$

$V = [0.2, 0.3]$

$V = [2, 0.3]$

$V = [2.2, 0.3]$

$V = [2.2, 0.3]$

$K = [1.1, 2.2]$

$K = [1, 3]$

$K = [1, 3]$

$K = [1.1, 3]$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

$$Q = \begin{bmatrix} 1.5 & 3 \\ 2 & 4 \\ 2.2 & 2.4 \\ 2.4 & 4.4 \end{bmatrix}$$

$$K = \begin{bmatrix} 1.1 & 2.2 \\ 1 & 3 \\ 1 & 3 \\ 1.1 & 3 \end{bmatrix}$$

$$V = \begin{bmatrix} 0.2 & 0.3 \\ 2 & 0.3 \\ 2.2 & 0.3 \\ 2.2 & 0.3 \end{bmatrix}$$

$$K^T = \begin{bmatrix} 1.1 & 1 & 1 & 1.1 \\ 2.2 & 3 & 3 & 3 \end{bmatrix}$$

↳ $Q \cdot K^T$:

$$\begin{bmatrix} 1.5 & 3 \\ 2 & 4 \\ 2.2 & 2.4 \\ 2.4 & 4.4 \end{bmatrix} \begin{bmatrix} 1.1 & 1 & 1 & 1.1 \\ 2.2 & 3 & 3 & 3 \end{bmatrix}$$

↳ dot product

$$Q_1 K_1 = (1.5 \times 1.1) + (3 \times 2.2) = 1.65 + 6.6 = 8.25$$

$$Q_1 K_2 = (1.5 \times 1) + (3 \times 3) = 1.5 + 9 = 10.5$$

$$Q_1 K_3 = (1.5 \times 1) + (3 \times 3) = 1.5 + 9 = 10.5$$

$$Q_1 K_4 = (1.5 \times 1.1) + (3 \times 3) = 1.65 + 9 = 10.65$$

$$Q_2 K_1 = (2 \times 1.1) + (4 \times 2.2) = 2.2 + 8.8 = 11$$

$$Q_2 K_2 = (2 \times 1) + (4 \times 3) = 2 + 12 = 14$$

$$Q_2 K_3 = (2 \times 1) + (4 \times 3) = 2 + 12 = 14$$

$$Q_2 K_4 = (2 \times 1.1) + (4 \times 3) = 2.2 + 12 = 14.2$$



$$Q_3 K_1 = (2.2 \times 1.1) + (4.4 \times 2.2) = 2.42 + 9.68 = 12.1$$

$$Q_3 K_2 = (2.2 \times 1) + (4.4 \times 3) = 2.2 + 13.2 = 15.4$$

$$Q_3 K_3 = (2.2 \times 1) + (4.4 \times 3) = 2.2 + 13.2 = 15.4$$

$$Q_3 K_4 = (2.2 \times 1.1) + (4.4 \times 3) = 2.42 + 13.2 = 15.62$$

$$Q_4 K_1 = (2.4 \times 1.1) + (4.4 \times 2.2) = 2.64 + 9.68 = 12.32$$

$$Q_4 K_2 = (2.4 \times 1) + (4.4 \times 3) = 2.4 + 13.2 = 15.6$$

$$Q_4 K_3 = (2.4 \times 1) + (4.4 \times 3) = 2.4 + 13.2 = 15.6$$

$$Q_4 K_4 = (2.4 \times 1.1) + (4.4 \times 3) = 2.64 + 13.2 = 15.84$$

$$\hookrightarrow QK^T, d_k = 2$$

$$\frac{1}{\sqrt{d_k}} \begin{bmatrix} 8.25 & 10.5 & 10.5 & 10.65 \\ 11 & 14 & 14 & 14.2 \\ 12.1 & 15.4 & 15.4 & 15.62 \\ 12.32 & 15.6 & 15.6 & 15.84 \end{bmatrix}$$

$$= \begin{bmatrix} 5.8336 & 7.4246 & 7.4246 & 7.5307 \\ 7.7782 & 9.8995 & 9.8995 & 10.0409 \\ 8.5560 & 10.889 & 10.8894 & 11.045 \\ 8.7116 & 11.0309 & 11.0309 & 11.2006 \end{bmatrix}$$

$$\hookrightarrow \text{softmax}:$$

$$= \begin{bmatrix} 0.0614 & 0.3016 & 0.3016 & 0.3354 \\ 0.0366 & 0.3056 & 0.3056 & 0.3521 \\ 0.0297 & 0.3063 & 0.3063 & 0.3578 \\ 0.0300 & 0.3046 & 0.3046 & 0.3609 \end{bmatrix} \times \begin{matrix} 0.2 & 0.3 \\ 2 & 0.3 \\ 2.2 & 0.3 \\ 2.2 & 0.3 \end{matrix}$$

$$\text{Attention}_{(Q,K,V)} = \begin{bmatrix} 2.0168 & 0.3 \\ 2.0656 & 0.3 \\ 2.0794 & 0.3 \\ 2.0792 & 0.3 \end{bmatrix}$$

	Wind	Humidity	Rain
1	weak	high	yes
2	strong	high	yes
3	weak	low	no
4	weak	low	yes
5	strong	low	no

Date _____ 20__

Q.3)

1)

Laplacian smoothing adds a constant α (usually 1) to avoid zero probability

Rain:

$$P(\text{Yes}) = \frac{3}{5} = 0.6$$

$$P(\text{No}) = \frac{2}{5} = 0.4$$

Wind - Conditional Probabilities

$$P(\text{Wind} = \text{Weak} \mid \text{Rain} = \text{Yes}) = \frac{2+1}{3+2} = 0.6$$

$$P(\text{Wind} = \text{Strong} \mid \text{Rain} = \text{Yes}) = \frac{1+1}{3+2} = 0.4$$

$$P(\text{Wind} = \text{Weak} \mid \text{Rain} = \text{No}) = \frac{1+1}{2+2} = 0.5$$

$$P(\text{Wind} = \text{Strong} \mid \text{Rain} = \text{No}) = \frac{1+1}{2+2} = 0.5$$

Humidity - Conditional Probabilities

$$P(\text{Humidity} = \text{High} \mid \text{Rain} = \text{Yes}) = \frac{2+1}{3+2} = \frac{3}{5} = 0.6$$

$$P(\text{Humidity} = \text{Low} \mid \text{Rain} = \text{Yes}) = \frac{1+1}{3+2} = \frac{2}{5} = 0.4$$

$$P(\text{Humidity} = \text{High} \mid \text{Rain} = \text{No}) = \frac{0+1}{2+2} = \frac{1}{4} = 0.25$$

$$P(\text{Humidity} = \text{Low} \mid \text{Rain} = \text{No}) = \frac{2+1}{2+2} = \frac{3}{4} = 0.75$$

↳ predicting weather combinations

$$P(\text{Class}) \times P(\text{Wind} \mid \text{Class}) \times P(\text{Humidity} \mid \text{Class})$$

↳ case 1 : wind = weak, humidity = high

$$\begin{aligned} \text{Score}(\text{Yes}) &= P(\text{Yes}) \times P(\text{Weak} \mid \text{Yes}) \times P(\text{High} \mid \text{Yes}) \\ &= 0.6 \times 0.6 \times 0.6 \\ &= 0.216 \end{aligned}$$

$$\begin{aligned} \text{Score}(\text{No}) &= P(\text{No}) \times P(\text{Weak} \mid \text{No}) \times P(\text{High} \mid \text{No}) \\ &= 0.4 \times 0.5 \times 0.25 \\ &= 0.050 \end{aligned}$$

$$\text{Sum} = 0.216 + 0.050 = 0.266$$

↳ normalised probabilities

$$P(\text{Yes} | \text{Weak, high}) = \frac{0.216}{0.266} = 0.812 \quad (81.2\%) \checkmark$$

$$P(\text{No} | \text{Weak, high}) = \frac{0.050}{0.266} = 0.188 \quad (18.8\%)$$

↳ Decision: Rain = Yes
(do not deliver)

↳ case 2: wind = weak, humidity = low

$$\text{Score (Yes)} = 0.6 \times 0.6 \times 0.4 = 0.144$$

$$\text{Score (No)} = 0.4 \times 0.5 \times 0.75 = 0.150$$

$$\text{Sum} = 0.144 + 0.150 = 0.294$$

$$\text{↳ normalize: } P(\text{Yes} | \text{Weak, Low}) = \frac{0.144}{0.294} = 0.49 \quad (49\%)$$

$$P(\text{No} | \text{Weak, Low}) = \frac{0.150}{0.294} = 0.510 \quad (51\%)$$

↳ decision: Rain = No
(do deliver)

↳ case 3: wind = strong, humidity = high

$$\text{Score (Yes)} = 0.6 \times 0.4 \times 0.6 = 0.144$$

$$\text{Score (No)} = 0.4 \times 0.5 \times 0.25 = 0.050$$

$$\text{sum} = 0.144 + 0.050 = 0.194$$

$$\text{↳ normalised: } P(\text{yes} | \text{strong, high}) = \frac{0.144}{0.194} = 0.742 \quad (74\%) \checkmark$$

$$P(\text{no} | \text{strong, high}) = \frac{0.050}{0.194} = 0.258 \quad (25.8\%)$$

↳ decision: Rain = Yes
(don't deliver)

↳ case 4: wind = strong, humidity = low

$$\text{↳ score (yes)} = 0.6 \times 0.4 \times 0.4 = 0.096$$

$$\text{↳ score (no)} = 0.4 \times 0.5 \times 0.75 = 0.15$$

$$\text{Sum} = 0.096 + 0.15 = 0.246$$

$$\text{↳ normalised: } P(\text{yes} | \text{strong, low}) = \frac{0.096}{0.246} = 0.39 \text{ (39\%)} \\ P(\text{no} | \text{strong, low}) = \frac{0.15}{0.246} = 0.61 \text{ (61\%)}$$

↳ decision: No Rain
(do deliver)

2) Information Gain & Entropy

↳ overall entropy $E(S)$ for Rain

$$E(S) = -\sum p_i \log_2(p_i) = -\left(\frac{3}{5}\right) \log_2\left(\frac{3}{5}\right) - \left(\frac{2}{5}\right) \log_2\left(\frac{2}{5}\right)$$

$$E(S) = -(0.6 \times -0.737) - (0.4 \times -1.322) \approx 0.442 + 0.529 = 0.971$$

↳ Information Gain for Wind

↳ Wind = weak

$$E(\text{Weak}) = -\left(\frac{2}{3}\right) \log_2\left(\frac{2}{3}\right) - \left(\frac{1}{3}\right) \log_2\left(\frac{1}{3}\right) = 0.918$$

↳ Wind = strong

$$E(\text{Strong}) = -\left(\frac{1}{2}\right) \log_2\left(\frac{1}{2}\right) - \left(\frac{1}{2}\right) \log_2\left(\frac{1}{2}\right) = 1.0$$

↳ Gain (Wind)

$$IG(\text{Wind}) = E(S) - \left[\left(\frac{3}{5}\right) E(\text{Weak}) + \left(\frac{2}{5}\right) E(\text{Strong}) \right]$$

$$IG(\text{Wind}) = 0.971 - [(0.6 \times 0.918) + (0.4 \times 1.0)] = 0.971 - [0.551 + 0.4]$$

$$= \boxed{0.020}$$

↳ Information Gain for Humidity

↳ Humidity = High

$$E(\text{High}) = 0$$

↳ ~~Low Humidity~~ = Low

$$E(\text{Low}) = -\left(\frac{1}{3}\right)\log_2\left(\frac{1}{3}\right) - \left(\frac{2}{3}\right)\log_2\left(\frac{2}{3}\right) \approx 0.918$$

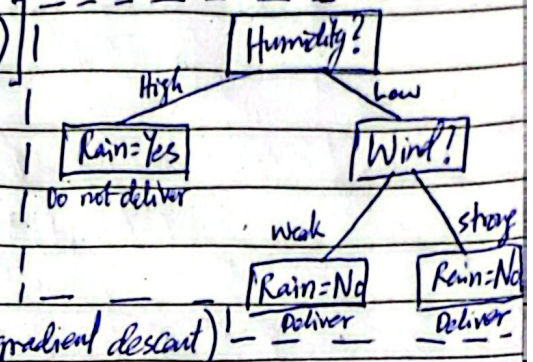
↳ Gain (Humidity)

$$IG(\text{Humidity}) = E(S) - \left[\left(\frac{2}{5}\right)E(\text{High}) + \left(\frac{3}{5}\right)E(\text{Low})\right]$$

$$= 0.971 - \left[(0.4 \times 0) + (0.6 \times 0.918)\right]$$

$$= 0.971 - 0.551$$

$$= 0.426$$



3) Logistic Regression Learning Algorithm

↳ relies on iterative optimization process (like gradient descent)

↳ data encoding:

$x_1(\text{Wind})$: weak=0, strong=1

$x_2(\text{Humidity})$: Low=0, High=1

$y(\text{Rain})$: No=0, Yes=1

↳ encoded dataset

1. $(x_1=0, x_2=1) \rightarrow y=1$

2. $(x_1=1, x_2=1) \rightarrow y=1$

3. $(x_1=0, x_2=0) \rightarrow y=0$

4. $(x_1=0, x_2=0) \rightarrow y=1$

5. $(x_1=1, x_2=0) \rightarrow y=0$

↳ setup of model:

↳ Hypothesis: $h_0(x) = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)}}$

↳ initialize weights to zero: $\theta_0=0, \theta_1=0, \theta_2=0$

↳ learning rate $\eta=0.1$

↳ initial predictions & errors ($e = h_0(x) - y$)

↳ sample 1: $e^{(1)} = 0.5 - 1 = -0.5$

↳ sample 2: $e^{(2)} = 0.5 - 1 = -0.5$

↳ sample 3: $e^{(3)} = 0.5 - 0 = 0.5$

↳ sample 4: $e^{(4)} = 0.5 - 1 = -0.5$

↳ sample 5: $e^{(5)} = 0.5 - 0 = 0.5$



↳ gradient: $\frac{1}{m} \sum (h_0(x^{(i)}) - y^{(i)}) x_j^{(i)}$

↳ gradient for θ_0 (Bias, $x_0=1$)

$$\frac{1}{5} [(-0.5) + (-0.5) + (0.5) + (-0.5) + (0.5)] = -0.1$$

↳ gradient for θ_1 (Wind)

$$\frac{1}{5} [(-0.5)(0) + (-0.5)(1) + (0.5)(0) + (0.5)(0) + (0.5)(1)] = 0$$

↳ gradient for θ_2 (Humidity)

$$\frac{1}{5} [(-0.5)(1) + (-0.5)(1) + (0.5)(0) + (-0.5)(0) + (0.5)(0)] = -0.2$$

↳ update weights

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \cdot \text{gradient}$$

$$\rightarrow \theta_0 = 0 - 0.1(-0.1) = 0.01$$

$$\rightarrow \theta_1 = 0 - 0.1(0.0) = 0.0$$

$$\rightarrow \theta_2 = 0 - 0.1(-0.2) = 0.02$$

↳ algo will repeat steps 3-5 w/ new weights to calc updated predictions & gradients converge.

4. Comparison of Learning Algorithms

↳ Naive Bayes (Generative)

↳ Assumes conditional independence between Wind & Humidity given Rain.

↳ uses Laplacian smoothing to eliminate zero probabilities, which makes it low-variance, fast and most robust choice for small datasets ($N=5$)

↳ Information Gain / Decision Trees (Non Parametric)

↳ Splits data based on entropy reduction, correctly identifying Humidity ($IG=0.420$) as the primary decision node over Wind ($IG=0.02$). It yields highly interpretable "if-then" delivery rules, but easily overfits small training sets.



↳ Logistic Regression (Discriminative)

↳ Directs model class probability using a sigmoid boundary without assuming feature independence. While it provides smooth, adjustable risk probabilities for delivery decisions, parameter optimization via gradient descent is sensitive to noise on very small datasets.

↳ Conclusion

↳ Naive Bayes is optimal when training data is sparse

↳ Information Gain provides ~~transparent~~ clear, easily verifiable decision rules.

↳ Logistic Regression outputs precise ~~into~~ probabilities so we can set custom percentage cutoffs.

Q.5)

1) The agent is a simple reflex agent, as it acts only on current percept

↳ the disease starts in each field with equal probability, Healthy & Infected are equally likely

$$P(A=I) = P(A=H) = \frac{1}{2} = 0.5$$

$$P(B=I) = P(B=H) = \frac{1}{2} = 0.5$$

↳ both fields are independent, so joint probability is product of individual probabilities.

$$P(A=I \cap B=I) = P(A=I) \times P(B=I) \\ = 0.5 \times 0.5$$

$$P(A \cap B) = 0.25$$

A	B	$P(A \cap B)$
Infected	Infected	$0.5 \times 0.5 = 0.25$
Infected	Healthy	$0.5 \times 0.5 = 0.25$
Healthy	Infected	$0.5 \times 0.5 = 0.25$
Healthy	Healthy	$0.5 \times 0.5 = 0.25$
		1.00

$$P(\text{both Infected}) = 0.25 \\ (25\%)$$



2) A & B initially infected, drone at A

Time	Location	State(A,B)	Percept	Action	Healthy fields	Credit	Total
0	A	(I,I)	[A, Infected]	Spray A	0	\$0	\$0
1	A	(H,I)	[A, Healthy]	Move to B	1	\$2	\$2
2	B	(H,I)	[B, Infected]	Spray B	1	\$2	\$4
3	B	(H,H)	[B, Healthy]	Move to A	2	\$4	\$8

Total Credit = \$8

Q.7) $A = (6, -1, \sqrt{9})$ $B = (4, X, -2)$

1) find X so that b is perpendicular to A

~~A.B~~ $A \cdot B = 0$ $\longrightarrow$ 2 vectors are perpendicular when dot prod is 0

$$(6)(4) + (-1)(X) + (3)(-2) = 0$$

$$24 - X - 6 = 0$$

$$18 - X = 0$$

$$\boxed{X = 18}$$

2)	Dot Product	Hadamard Product
symbol :	$a \cdot b$	$a \odot b$
operation :	multiply corresponding matching elements & sum	multiply matching elements & store separate
formula :	$a \cdot b = a_1b_1 + a_2b_2 + \dots + a_nb_n$	$a \odot b = (a_1b_1, a_2b_2, \dots, a_nb_n)$
output :	scalar / (single number)	vector of same size
geometric meaning :	$ a b \cos\theta \rightarrow$ measure of similarity	none directly
used in :	Attention Scores (Q.K), cosine similarity b/w embeddings	LSTM/GRU gates, masking,

3) prove $u \cdot v = 6$ $u: (1, 2)$ $v: (2, 2)$

↳ using L2 Norm

↳ add vectors: $u + v = (1+2, 2+2)$
 $u + v = (3, 4)$

↳ squared L2 norms.

$$\|u\|^2 = 1^2 + 2^2 = 5$$

$$\|v\|^2 = 2^2 + 2^2 = 8$$

$$\|u+v\|^2 = 3^2 + 4^2 = \sqrt{25} = 25$$

↳ identity: $\|u+v\|^2 = \|u\|^2 + \|v\|^2 + 2(u \cdot v)$

$$25 = 5 + 8 + 2(u \cdot v)$$

$$25 = 13 + 2(u \cdot v)$$

$$12 = 2(u \cdot v)$$

$$\boxed{6 = (u \cdot v)} \text{ proved}$$

↳ using vector projection

↳ find u_{proj} - pt where perpendicular from u meets the line of v

↳ v lies on line $y = x$, slope = 1

↳ perpendicular thru $u = (1, 2)$ has slope = -1 $\rightarrow y = -x + 3$

↳ two lines meet where: $x = -x + 3$

$$x = \frac{3}{2}$$

↳ so $u_{\text{proj}} = (\frac{3}{2}, \frac{3}{2})$

↳ lengths.

$$\|u_{\text{proj}}\| = \sqrt{1.5^2 + 1.5^2} = \sqrt{4.5} = \frac{3\sqrt{2}}{2}$$

$$\|v\| = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

↳ dot product = proj length times length of v

$$u \cdot v = \|u_{\text{proj}}\| \times \|v\|$$

$$u \cdot v = (1.5\sqrt{2}) \times (2\sqrt{2})$$

$$\boxed{u \cdot v = 6} \text{ proved}$$



Q-8) $S(t) = A(t) + B(t) + C(t) + D(t)$

$$S(t) = (0.2t + t^2) + (\sin t + \cos t) + (t^5 + t^6 + \sin t) + (t^5 + t^8 + \sin t + \cos t)$$

$$S(t) = 0.2t + t^2 + 2t^5 + t^6 + t^8 + 3\sin t + 2\cos t$$

$t=1:$

$$S(1) = 0.2(1) + (1)^2 + 2(1)^5 + (1)^6 + (1)^8 + 3\sin(1) + 2\cos(1)$$

$$S(1) = 7.252$$

$t=2:$

$$S(2) = 0.2(2) + (2)^2 + 2(2)^5 + (2)^6 + (2)^8 + 3\sin(2) + 2\cos(2)$$

$$S(2) = 390.5035$$

$t=3:$

$$S(3) = 0.2(3) + (3)^2 + 2(3)^5 + (3)^6 + (3)^8 + 3\sin(3) + 2\cos(3)$$

$$S(3) = 7787.7543$$

$t=4:$

$$S(4) = 0.2(4) + (4)^2 + 2(4)^5 + (4)^6 + (4)^8 + 3\sin(4) + 2\cos(4)$$

$$S(4) = 71699.0044$$

